

Exercise 6

For $z \in \mathbb{R}^n$, write $\hat{F}_z(v) = \frac{1}{n} \sum_{i=1}^n \chi_{\{z_i \leq v\}}$ for the empirical cumulative distribution function of z with $\hat{F}_z(-\infty) = 0$ and $\hat{F}_z(+\infty) = 1$. Write

$$\text{Quantile}(z; \tau) = \inf \left\{ v \in \mathbb{R} \mid \hat{F}_z(v) \geq \tau \right\}$$

for the empirical τ -quantile of z for any $\tau \in [0, 1]$. Consequently, $\text{Quantile}(z; 0) = -\infty$.

Prove that, for any $z \in \mathbb{R}^n$, the following statements are true:

1. $\hat{F}_z(v) = \sup \{ \tau \in [0, 1] \mid \text{Quantile}(z; \tau) \leq v \}$ for all $v \in \mathbb{R}$.
2. $\text{Quantile}(z; \hat{F}_z(v)) \leq v$ for all $v \in \mathbb{R}$.
3. $\hat{F}_z(\text{Quantile}(z; \tau)) \geq \tau$ for all $\tau \in [0, 1]$.
4. If z has no repeated values, then $\hat{F}_z(\text{Quantile}(z; \tau)) = \frac{\lfloor n\tau \rfloor}{n}$ for all $\tau \in [0, 1]$.

Solution Let $z \in \mathbb{R}^n$ be given and fixed. Let us introduce a notation S_τ for the set of all $v \in \mathbb{R}$ such that $\hat{F}_z(v) \geq \tau$. Then, by definition, $\text{Quantile}(z; \tau) = \inf S_\tau$. All four statements depend on the following lemma:

Lemma 1. If $v \in \mathbb{R}$ and $\tau \in [0, 1]$, then $\text{Quantile}(z; \tau) \leq v$ if and only if $\tau \leq \hat{F}_z(v)$.

Proof. If $\tau \leq \hat{F}_z(v)$, then $v \in S_\tau$ and $v \geq \inf S_\tau = \text{Quantile}(z; \tau)$.

Let us focus on the converse. Suppose that $\text{Quantile}(z; \tau) \leq v$. For simplicity, put $q = \text{Quantile}(z; \tau)$. If $\tau = 0$, then $\hat{F}_z(v) \geq 0 = \tau$ holds trivially. Assume from now on that $\tau > 0$. Then q is finite number. Choose $\delta > 0$ small enough such that no points z_i lies between q and $q + \delta$, which is always doable because z has only finitely many points. Since q is the infimum, choose $s \in S_\tau$ such that $q \leq s < q + \delta$. Then, we arrive at

$$\hat{F}_z(v) \geq \hat{F}_z(q) = \hat{F}_z(s) \geq \tau$$

by applying in the first inequality the monotonicity of $\hat{F}_z$ and in the second equality the fact that no points z_i lies between q and $q + \delta$. The lemma is proved. $\square$

We now prove the four statements. To prove the first statement, let $v \in \mathbb{R}$ be given and we have, by the lemma:

$$\{ \tau \in [0, 1] \mid \text{Quantile}(z; \tau) \leq v \} = \left\{ \tau \in [0, 1] \mid \tau \leq \hat{F}_z(v) \right\} = [0, \hat{F}_z(v)]$$

and the supremum of the interval is just $\hat{F}_z(v)$. The second and the third statements are already included in the lemma. The fourth statement is trivial given the special assumption that z has no repeated values.////